Polymers, plastics and fibre-reinforced composites

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General

- Plastics consist of long chain molecules, called polymers. Polymers have a large number of repetitive units called monomers.
- Monomers are organic material (C & H). Polymerisation is a process of joining monomers.
- Their behaviour is highly dependent on the arrangement of the chains and the chemical composition;
- Plastics can
  - flexible & stretchable, e.g. polyethylene;
  - brittle, e.g., perspex, polystyrene;
  - transparent, e.g., perspex and polycarbonate;
  - tough, e.g., nylon

Polymers

Advantages

- Low density (lightweight)
- Low thermal conductivity (as insulation at room temperature)
- Easily moulded/shaped
- Good resistance to chemicals and not subject to electrochemical corrosion

Disadvantages

- A low modulus (often less than 2% of steel), hence, they are unsuitable for use as structural materials.
- Higher thermal expansion than other material, value ranges from 50-300x10^{-6}/ºC (compare with that of steel or concrete 10x10^{-6}/ºC)
- Poor resistance to heat, may be sensitive to ignition or may release toxic fumes
- Degradation due to ultra-violet light is common. It can result in brittleness, increased opacity of a transparent material, loss in strength, cracking, etc.

Types of polymers

Thermosetting polymers, e.g., epoxy, urea, polyester, and polyurethane.

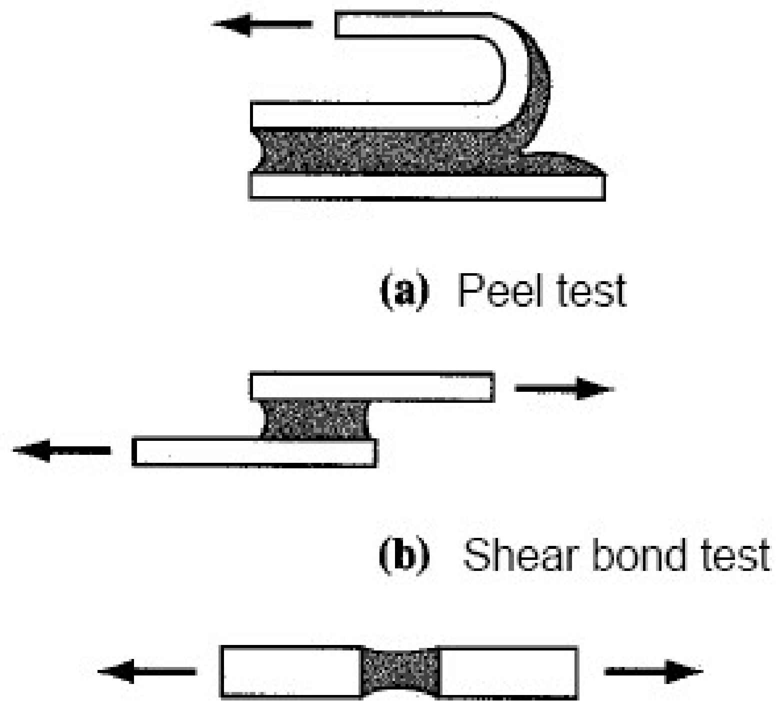
- Dense cross-linking between neighbouring chains, results in hard, rigid and inflexible material with low toughness.
- On application of heat, it starts decomposing before melting .

Thermoplastic polymers, e.g., polyethylene, polyvinyl chloride (PVC), teflon (PTFE).

- Can be melted (repeatedly) at 100ºC-150ºC so they can be easily formed and reformed (re-cycled).
- Modulus is very low as the neighbouring chains are held by van der waal bonds.
- It can very tough (before the plastic can break it has to unfold and lots of energy is expended in this unravelling).
- Each thermoplastic has a characteristic temperature called the glass transition temperature, T_g , which defines the boundary temperature between more rigid (E higher), brittle behaviour and more flexible (E lower), ductile behaviour. It is analogous to the ductile-brittle transition temperature in steel.

Schematic description of bond tests

- One of the common applications of polymeric material is as coating, sealant or adhesive where bond properties is important.
- Various tests are needed to determine the various mode of action:



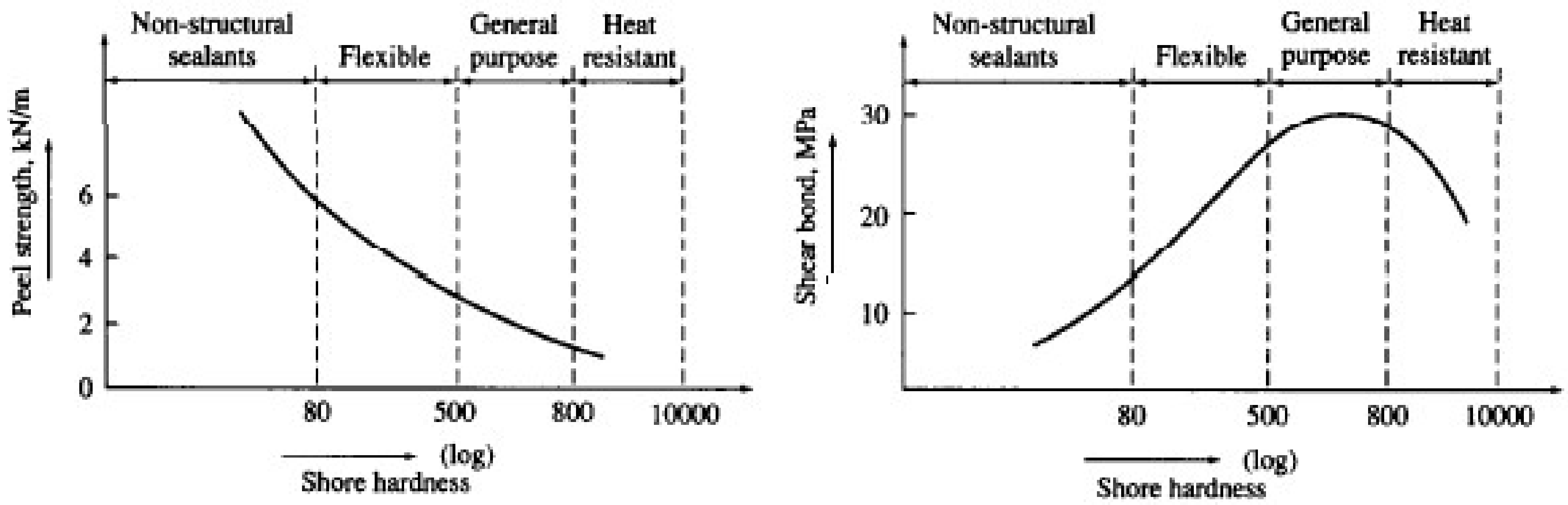
(a) Peel test

(b) Shear bond test

(c) Tensile bond test

Source : Young et al (1998)

Classification of adhesives and sealants based on bond strength and hardness



Source : Young et al (1998)

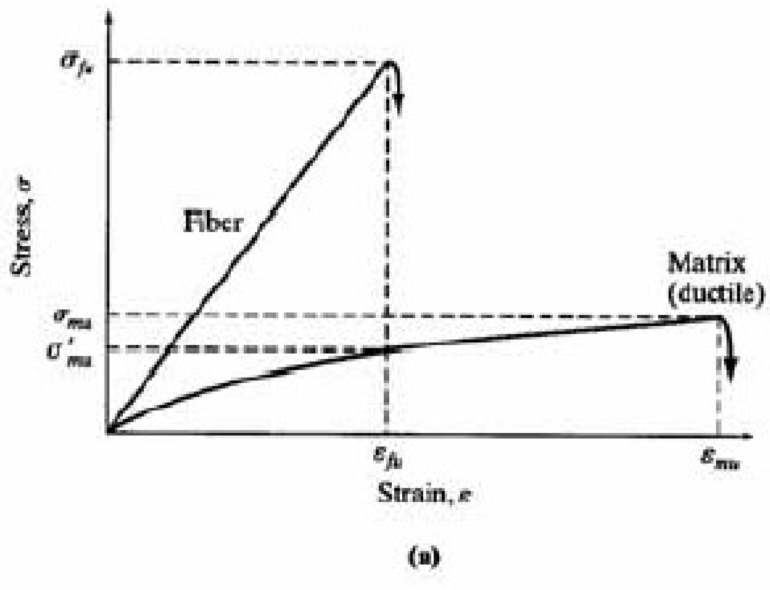
Fibre-reinforced composites

- Composite is a heterogeneous mixture of two or more homogenous phases bonded together

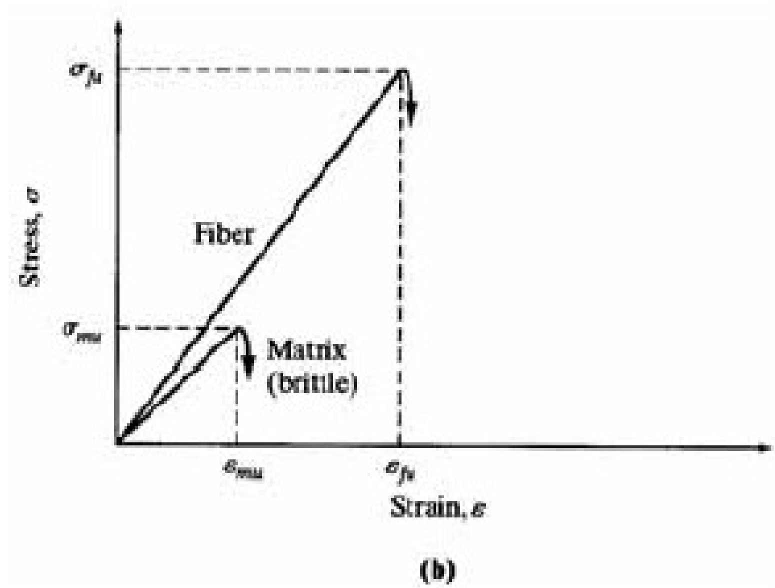
- Matrix can be cementitious or polymeric
  - Cementitious matrix – Fibre-reinforced Concretes (FRC)
  - Polymeric matrix – Fibre-reinforced Polymers (FRP)

Fibres are embeded in the matrix. Each fibre has a small cross-sectional area with least flaws & defects

Stress-strain curves of components in fibre-reinforced composites



fibre -ductile matrix system
[characteristics of FRP]



fibre -brittle matrix systems
[characteristics of FRC]

Fibre-reinforced composites

Fibre-reinforced concrete (FRC)

- Due to low fluidity & higher viscosity, fibres cannot be mixed properly with the matrix
- Fibre content is less than about 15% without aggregate, and less than about 2% in the presence of aggregate
- Too much fibre in the mix leads to lower workability, and clumping of fibres
- Addition of fibre increases toughness and strength of the element

Fibre-reinforced polymer (FRP)

- High fluidity & low viscosity, hence fibre content is relatively higher (50 - 80%)
- Since the matrix is ductile while the fibre is strong & rigid the resulting FRP composite will be more ductile than FRC

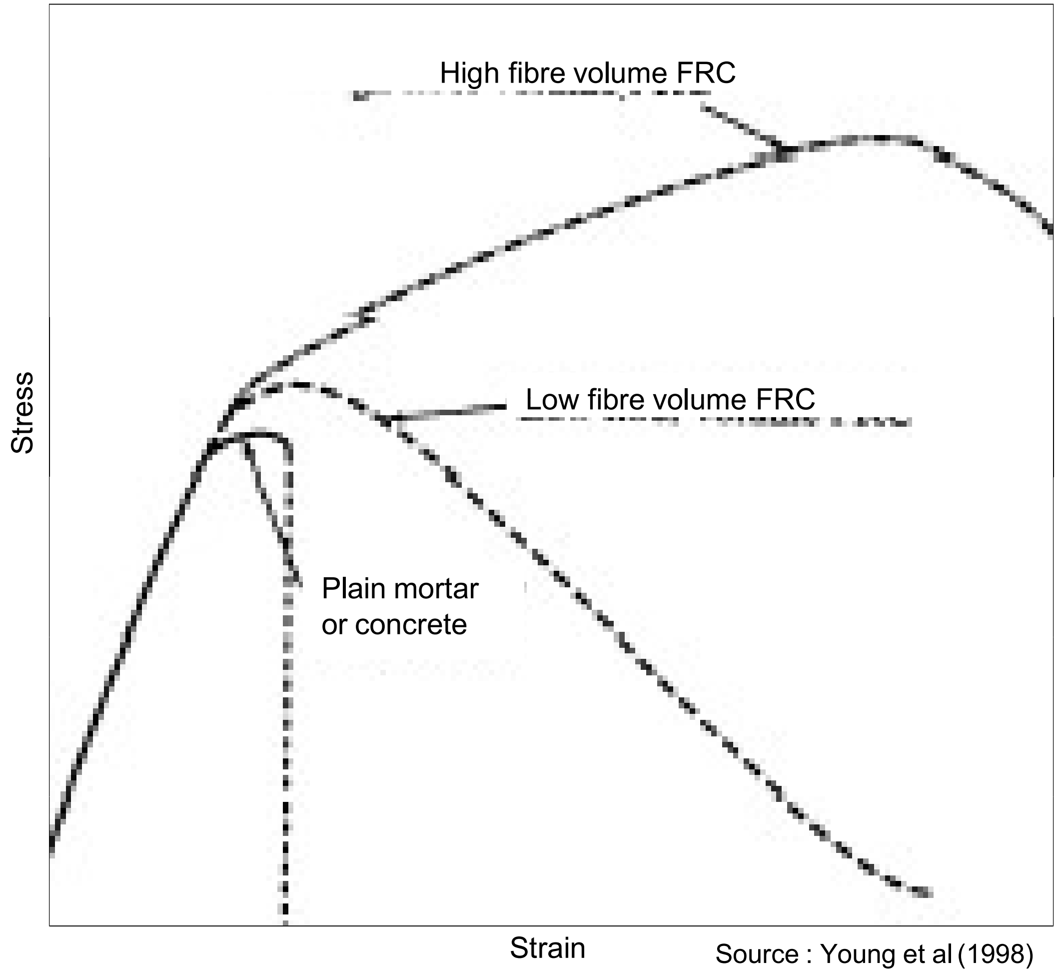
Fibre-reinforced concrete (FRC)

TABLE 16.1 Typical Properties of Fibers

Fiber	Diameter (μm)	Relative Density[a]	Modulus of Elasticity (GPa)	Tensile Strength (GPa)	Elongation at Break (%)
Steel	5–500	7.84	200	0.5–2.0	0.5–3.5
Glass	9–15	2.60	70–80	2–4	2–3.5
Asbestos					
Crocidolite	0.02–0.4	3.4	196	3.5	2.0–3.0
Chrysotile	0.02–0.4	2.6	164	3.1	2.0–3.0
Fibrillated polypropylene	20–200	0.9	5–77	0.5–0.75	8.0
Aramid (Kevlar)	10	1.45	65–133	3.6	2.1–4.0
Carbon (high strength)	9	1.90	230	2.6	1.0
Nylon	—	1.1	4.0	0.9	13.0–15.0
Cellulose	—	1.2	10	0.3–0.5	—
Acrylic	18	1.18	14–19.5	0.4–1.0	3
Polyethylene	—	0.95	0.3	0.7×10^{-3}	10
Wood Fiber	—	1.5	71.0	0.9	—
Sisal	10–50	1.50	—	0.8	3.0

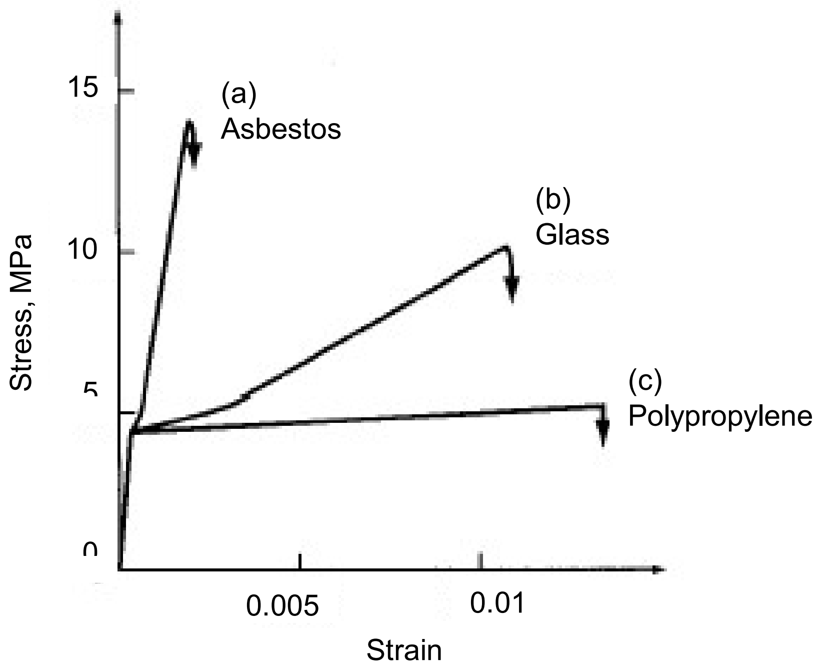
[a]Formerly specific gravity.

Schematic stress-strain curves for low and high fibre volume FRC



Source : Young et al (1998)

Stress-strain behaviour of three types of FRC



Source : Young et al (1998)

a) Asbestos fibres: strong bond, many fine cracks closely spaced, strong but limited ductility and ability to withstand impact.

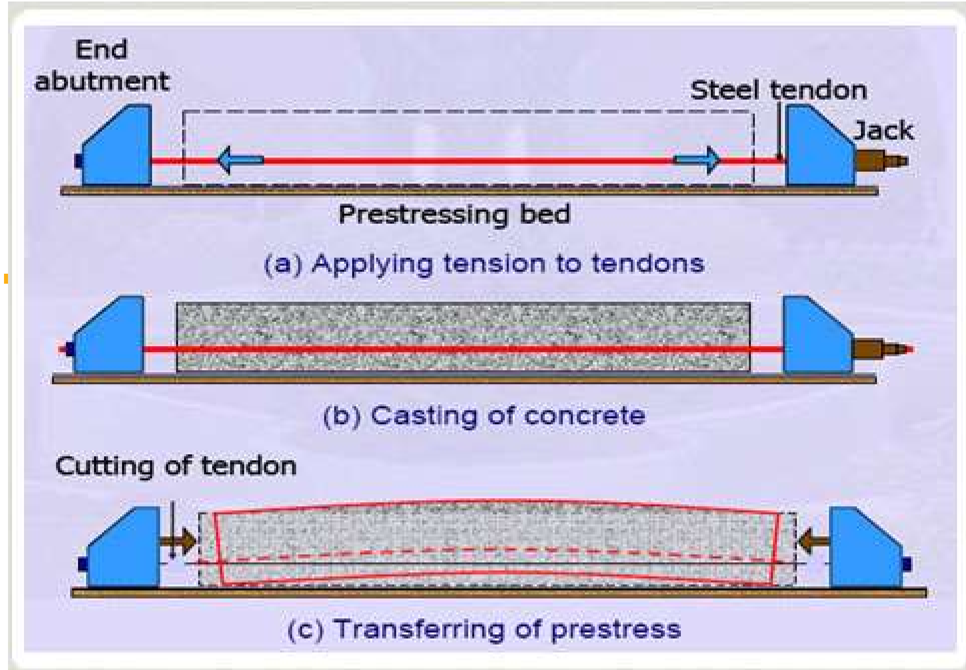
a) Glass fibres : moderate bond, multiple cracking but greater spacing, more pullout of fibre s: low bond and low E; and thus cracks are widely spaced.
Fibres debond with considerable pullout. High ductility and resistance to impact.

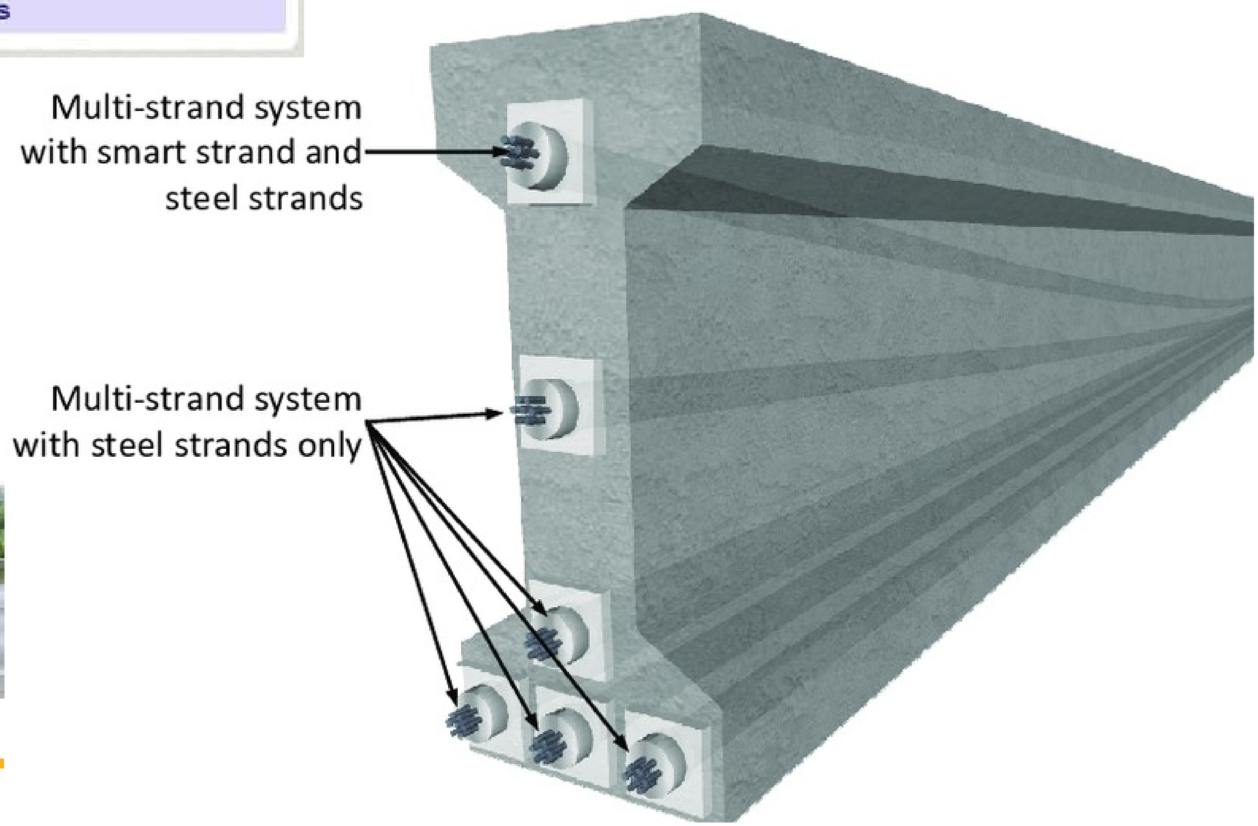
a) Polyproplylene fibres: low bond and low E; and thus cracks are very widely spaced. Fibres debond with considerable pullout. High ductility and resistance to impact

Fibre-reinforced polymer (FRP)

https://www.youtube.com/watch?v=_84SFqZHvjA

https://www.youtube.com/watch?v=OIHwGP85R1o

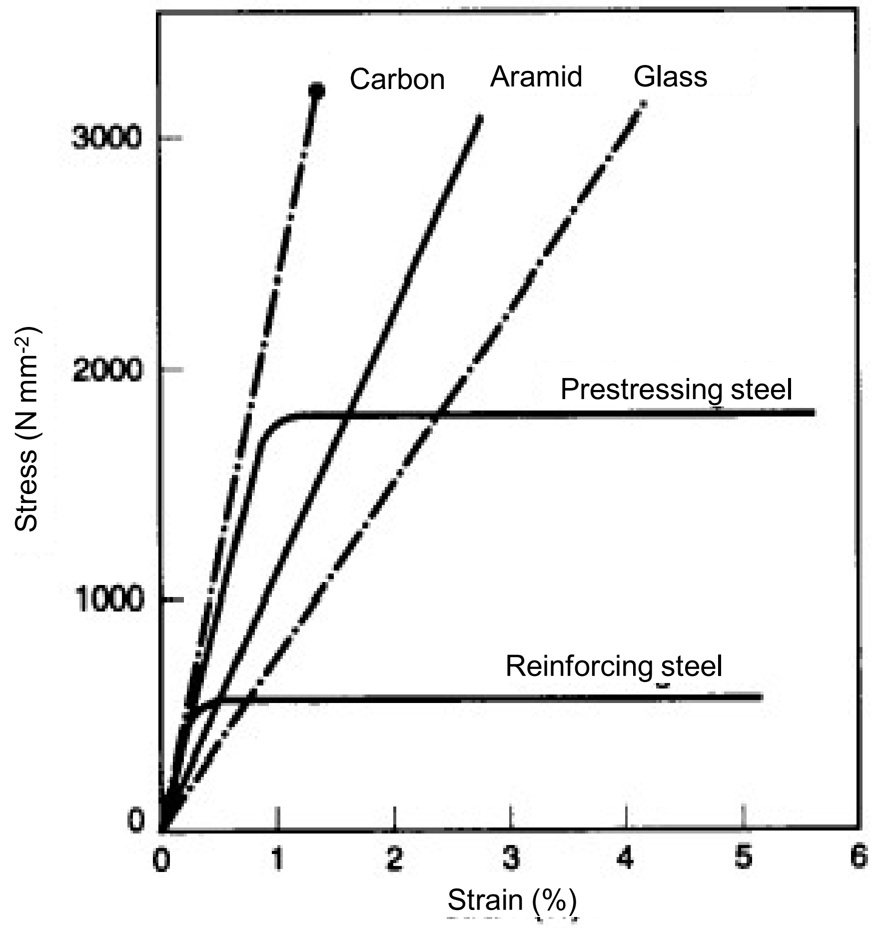

End
abutment
Steel tendon
Jack
Prestressing bed
(a) Applying tension to tendons
(b) Casting of concrete
Cutting of tendon
(c) Transferring of prestress


Multi-strand system
with smart strand and
steel strands
Multi-strand system
with steel strands only

Typical mechanical properties of a range of reinforcing fibres

	E-Glass	Aramid (Kevlar49)	High strength carbon	High modulus carbon	Steel (Fe 360)
Tensile strength (N/mm^2)	2400	3600	3300 - 6370	2600 - 4700	235 (yield) 360 (ultimate)
Tensile modulus (kN/mm^2)	70	130	230 - 300	345 - 590	205
Failure strain (%)	3.5	2.5	1.5 - 2.2	0.6 - 1.4	20
Density (kg/m^3)	256 0	1440	1800	1900	7850

Stress-strain relations for engineering fibres and reinforcements



Source : Jackson and Dhir 1996

Characteristics of various fibres

E-Glass

- Individual fibres or filaments are produced with diameters around 10 μm
- Lower strength and stiffness than the other fibres being considered, but is also considerably lower in cost.
- Electrically non-conductive and generally exhibits good chemical resistance, but may require protection against strong acid or alkali environments.
- Chopped fibre mats are mainly used for non-structural and secondary structure applications and produce laminates with relatively low strength.

Aramid

- Very high tensile strengths and low density.
- Aramid fabrics are very soft and easy to handle but can prove difficult to cut with conventional tools due to their very high toughness.
- Negative coefficient of thermal expansion.
- Resistant to most forms of chemical attack.
- Aramid changes colour under UV and strength can be reduced.

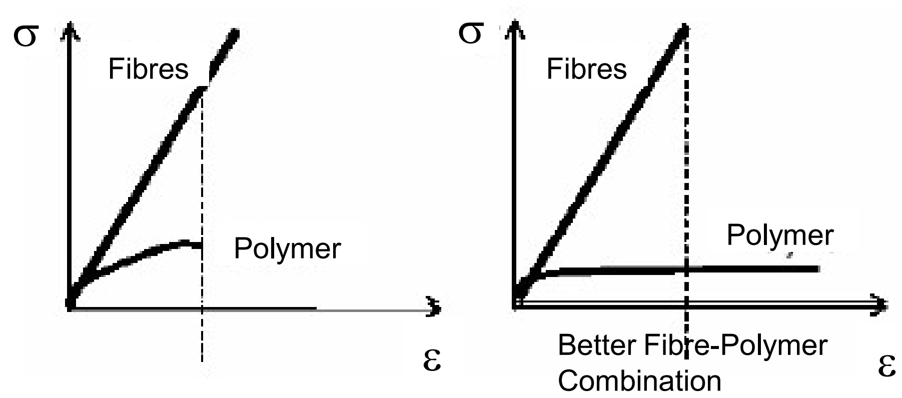
Characteristics of various fibres

Carbon

- Very high strength and stiffness is the most commonly used fibre in Aerospace.
- Generally classified as either high strength or high modulus, although in reality both the strength and modulus of most carbon fibres are very high.
- Negative coefficient of thermal expansion.
- Resistant to most forms of chemical attack.
- Electrically conductive. In many applications this may not be relevant but may be critical in others, e.g. example near railway overhead power lines.
- Exhibits a very high electrolytic potential and while it will not corrode itself, it may cause galvanic corrosion in other materials.

The polymer matrix

- It binds the fibres together, supporting and protecting them.
- It has far lower strength than the fibres. Rather than carrying the load, the polymer matrix transfers load to the fibres through shear stresses at the fibrematrix interface.
- A stronger thermosetting polymer is typically used for the matrix material in FRP. The three main types are Polyester, Epoxy and Vinyl Ester.
- To utilize the full strength of the fibres, a polymer matrix that has a rupture strain greater than the rupture strain of the fibres should be selected.
- Direct exposure to sunlight embrittles all resin

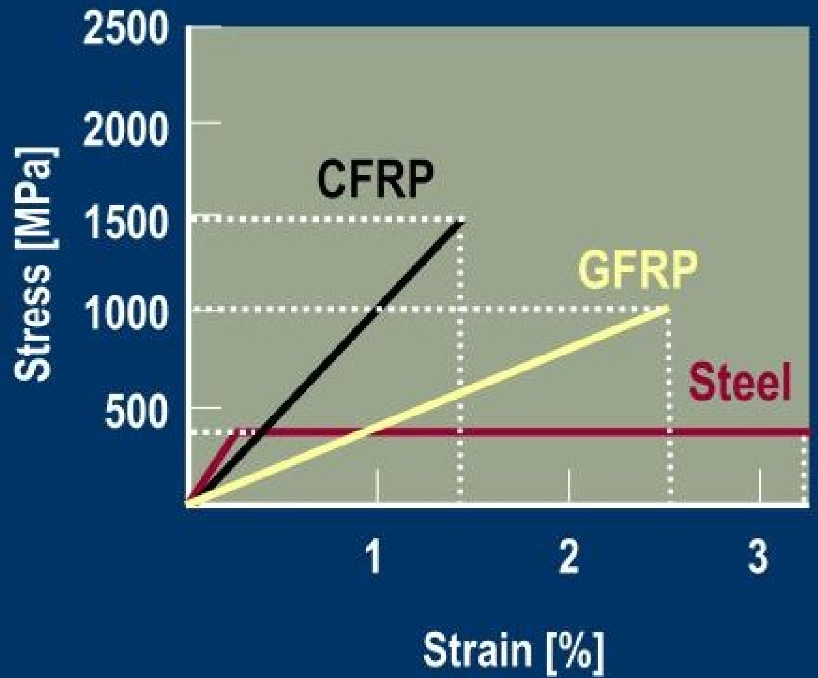


Mechanical properties of FRP

- The strength and stiffness of the composite lie between those of the fibre and the matrix and depend upon the percentage of each component.
- However, the total force resisted by FRP depends mainly on the cross-sectional area of unidirectional fibres provided. The matrix doesn't carry much force, but enables the fibres to reach their full potential load carrying capacity.
- FRP materials are linearly elastic until failure. The brittle failure is sudden and occurs with almost no warning. This lack of permanent deformation before failure is a major difference between FRP and steel and requires consideration during design of the structural member.
- Creep is mainly a function of the polymer matrix not the fibres. For FRP with unidirectional fibres, where the matrix contributes little to the mechanical properties of the composite, the effect of creep is usually not a concern.

Mechanical Properties

- FRP properties
  (in general versus steel):
  - Linear elastic behaviour to failure
  - No yielding
  - Higher ultimate strength
  - Lower strain at failure
  - Comparable modulus (carbon FRP)

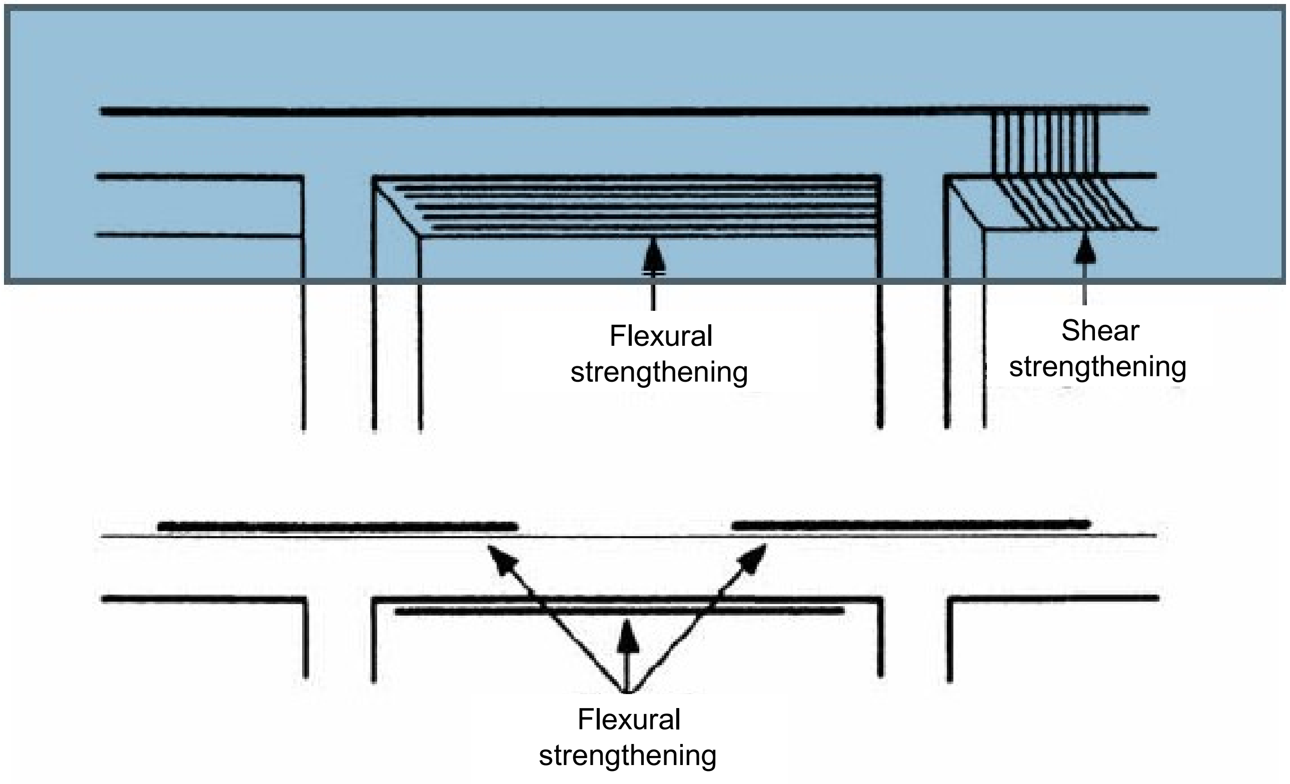


Structural retrofitting using composites

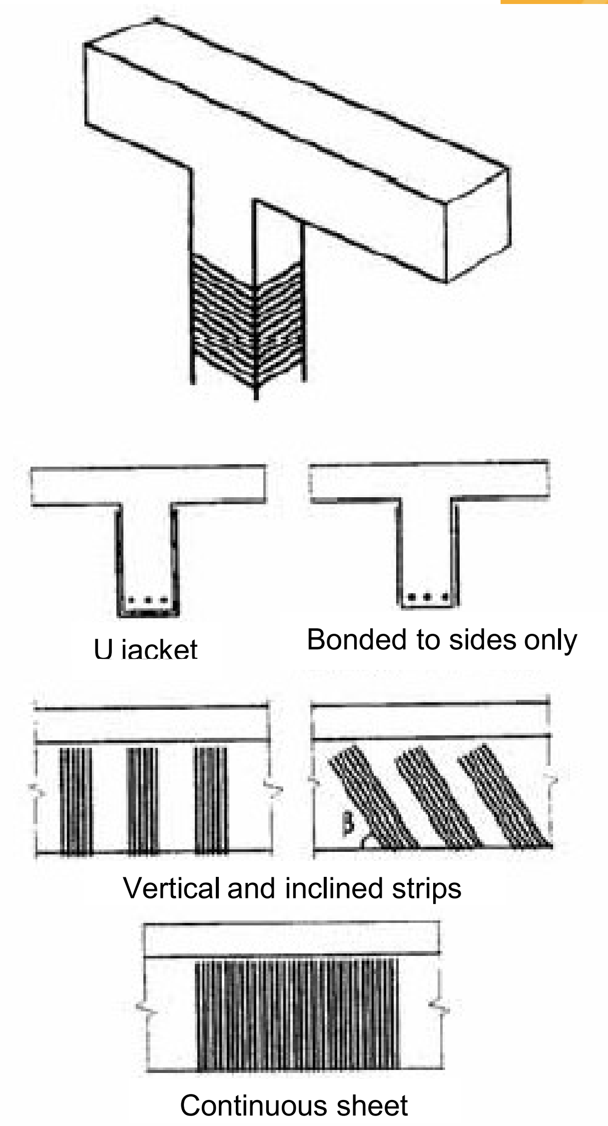
Examples of FRP applications in civil engineering

- To increase the flexural strength of reinforced concrete slabs.
- To increase the shear or flexural strength of reinforced concrete beams.
- To increase the strength and ductility of reinforced concrete columns.
- To increase the blast resistance of walls.
- To repair and increase the weathering resistance of concrete structures e.g. chimney , tank walls.

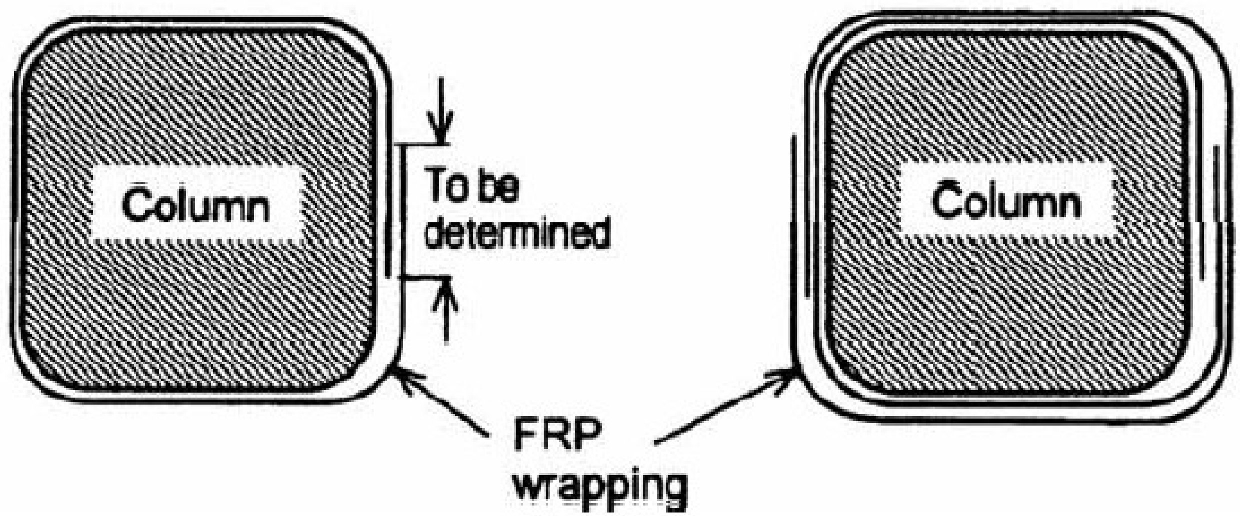
Various ways of strengthening using FRP



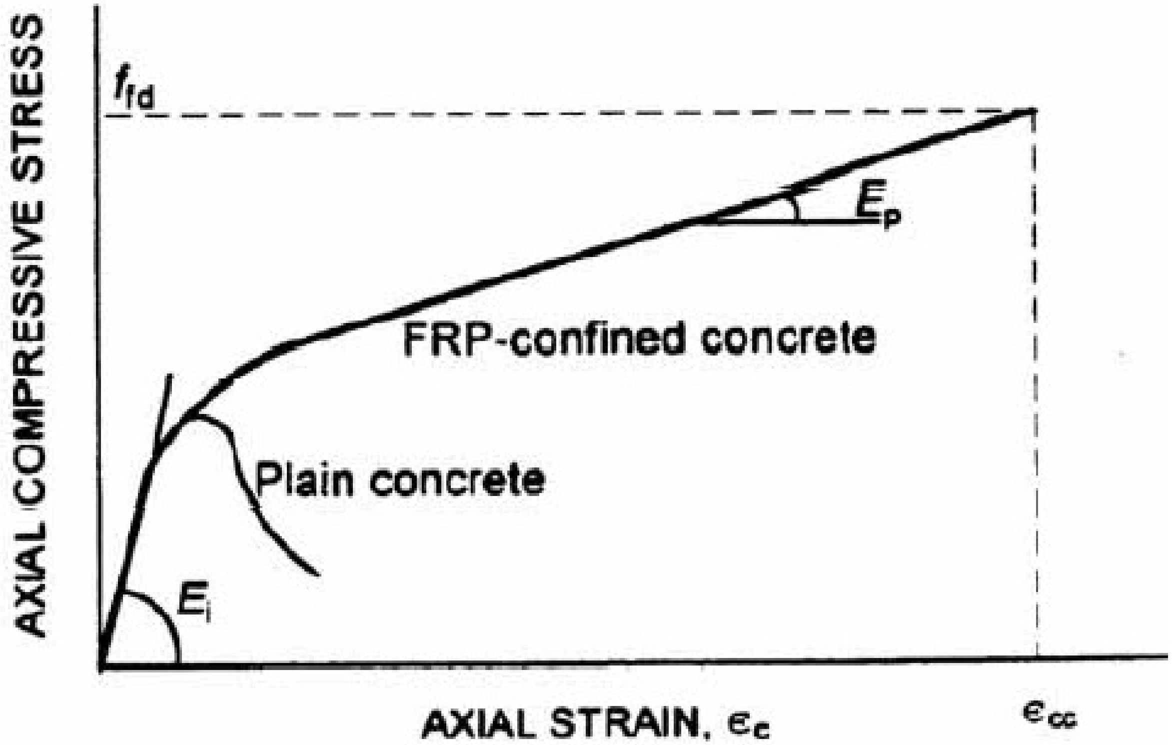
Shear strengthening



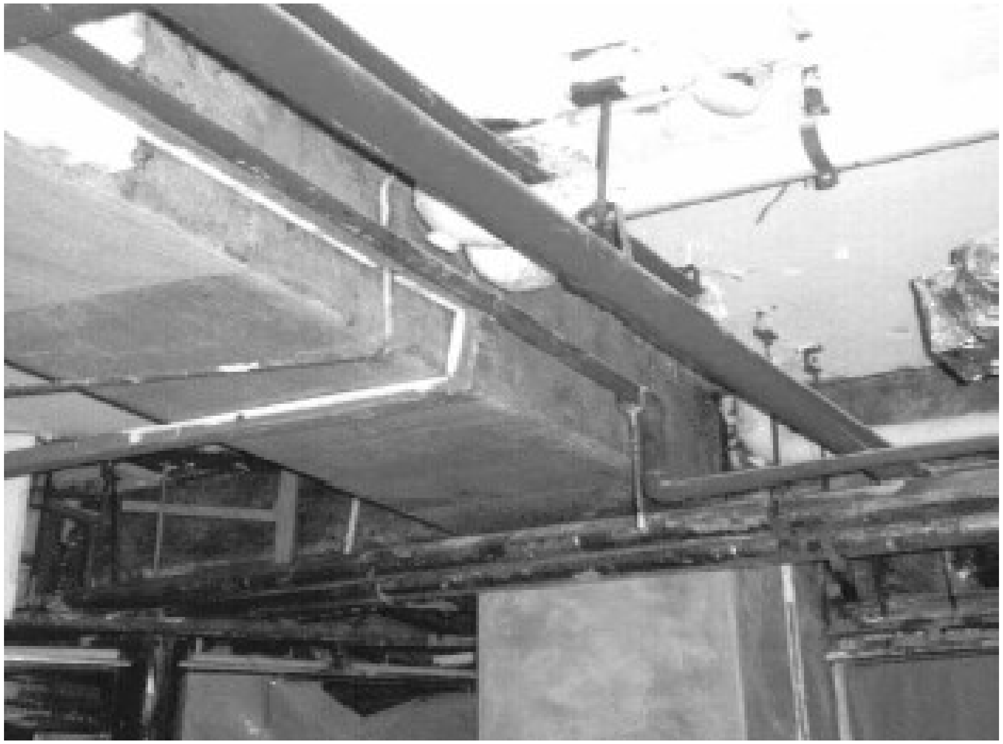
Strengthening of columns



Enhancement of axial behaviour from lateral confinement



International Building, Singapore



Beam upgrading completed without disruption to piping and mechanical services

Hotel Rendezvous, Singapore

Application to beams for flexural and shear enhancement

Hotel Rendezvous, Singapore

Strengthened column and beams

Hotel Rendezvous, Singapore

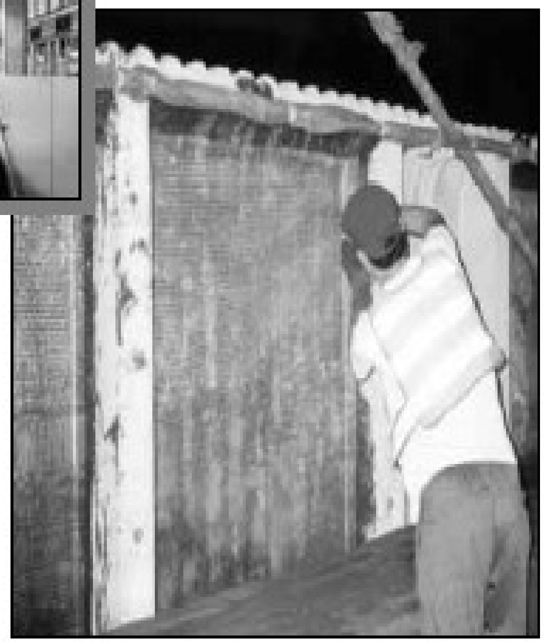
3) Walls strengthening

Shah House, Mumbai-India

Columns and Flat slab strengthened
with FRP strip system

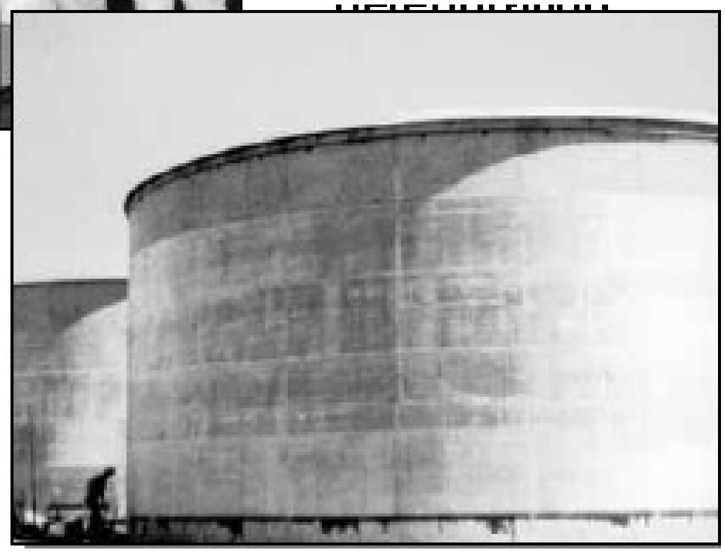
MTRC, Hong Kong

FRP strengthening of barrier walls due to cracking caused by vibrations



Fort Lewis, USA

Application of the FRP system to contain hoop stresses and prevent further Deterioration



All FRP structure

Applications

FRP road bridge

Laboratory testing of an FRP bridge deck panel

FRP reinforced concrete

Applications

Case study 1: Taylor Bridge

Headingley, Manitoba
Opened 1998
165.1 metre span
2-lanes

FRP reinforced concrete

Wotton Bridge

Placement of glass FRP deck reinforcement